

Wazuh + Suricata IDS Integration

Network Intrusion Detection & Auto-Block

Complete Step-by-Step Setup Guide

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Overview

This guide covers the complete setup of Suricata IDS integrated with Wazuh SIEM to automatically detect network attacks and block malicious IPs. Written from a real working environment — every command was tested and confirmed working.

What this setup does:

- Suricata monitors all network traffic in real-time on the agent machine
- When a suspicious packet matches a rule (50,861 rules loaded), Suricata writes an alert to eve.json
- Wazuh Agent reads eve.json and forwards alerts to Wazuh Manager
- Custom rule 86701 fires at Level 7 for any Suricata alert
- Active Response runs suricata-drop.sh — the attacker's IP gets blocked via iptables automatically
- Alert is visible in Wazuh Dashboard under Security Events

Environment used:

Component	Details
Wazuh Manager	Ubuntu Server 22.04 (hostname: wazuh-soc, IP: 103.228.38.219)
Wazuh Agent	Ubuntu Desktop 22.04 (hostname: rakib-virtual-machine, IP: 192.168.79.131)
Wazuh Version	4.14.5
Suricata Version	8.0.5
Network Interface	ens33

Step 1 — Install Suricata on the Agent

NOTE: Run all commands in this step on: AGENT (rakib-virtual-machine)

Add the official Suricata stable repository:

```
sudo add-apt-repository ppa:oisf/suricata-stable -y
sudo apt update
```

If suricata-update package conflict appears, remove it first:

```
sudo apt remove suricata-update -y
```

WARNING: Suricata 8.x includes suricata-update built-in. The old standalone package conflicts during upgrade. Always remove it before installing 8.x.

Install Suricata:

```
sudo apt install suricata -y
# Verify version
suricata --version
# Expected: This is Suricata version 8.0.5 RELEASE
```

Find your network interface:

```
ip a
# Look for the interface with your IP – in this setup: ens33
```

Step 2 — Configure Suricata

NOTE: Run on: AGENT (rakib-virtual-machine)

Download detection rules (Emerging Threats ruleset):

```
sudo suricata-update
# This downloads 50,000+ attack detection rules
# Rules saved to: /var/lib/suricata/rules/
```

Edit the main Suricata config file:

```
sudo nano /etc/suricata/suricata.yaml
```

Set your network interface (search with Ctrl+W for "af-packet"):

```
af-packet:
  - interface: ens33    # replace with your interface name

pcap:
  - interface: ens33    # same here
```

Ensure eve.json output is enabled (search for "eve-log"):

```
- eve-log:
  enabled: yes
  filetype: regular
  filename: eve.json
  types:
    - alert
    - http
    - dns
    - tls
    - flow
```

Validate config and start Suricata:

```
# Test config – look for "Configuration provided was successfully loaded"
sudo suricata -T -c /etc/suricata/suricata.yaml -v

# Enable and start
sudo systemctl enable suricata
sudo systemctl start suricata
sudo systemctl status suricata
# Expected: Active: active (running)
```

Verify eve.json is being written:

```
sudo tail -f /var/log/suricata/eve.json
```

```
# You should see JSON lines with event_type: stats, dns, etc.
```

```
# Press Ctrl+C to exit
```

Step 3 — Fix Permissions for Wazuh Agent

NOTE: Run on: AGENT (rakib-virtual-machine)

By default, /var/log/suricata/ is owned by the suricata group and is not readable by the wazuh user. This must be fixed or Wazuh will silently fail to read eve.json.

Check current permissions:

```
ls -la /var/log/suricata/  
# You will see: drwxrwx--- suricata suricata  
# The wazuh user cannot enter this directory
```

Add wazuh user to the suricata group:

```
sudo usermod -aG suricata wazuh
```

Restart Wazuh agent to apply group change:

```
sudo systemctl restart wazuh-agent
```

Verify wazuh can now read eve.json:

```
sudo -u wazuh cat /var/log/suricata/eve.json | head -3  
# Should output JSON – if Permission denied appears, recheck the usermod step
```

Step 4 — Configure Wazuh Agent to Monitor eve.json

NOTE: Run on: AGENT (rakib-virtual-machine)

Edit the agent ossec.conf:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add this block before :

```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
```

NOTE: *log_format json* tells Wazuh to parse each line as JSON. This is required for the Suricata rules to match correctly.

Restart the Wazuh agent:

```
sudo systemctl restart wazuh-agent
sudo systemctl status wazuh-agent
# Expected: Active: active (running)
```

Verify Wazuh is reading the file:

```
sudo grep "eve.json" /var/ossec/logs/ossec.log
# Expected: INFO: (1950): Analyzing file: '/var/log/suricata/eve.json'
```

Step 5 — Add Custom Suricata Rule on the Manager

NOTE: Run on: MANAGER (wazuh-soc)

Wazuh has a built-in Suricata rule file at `/var/ossec/ruleset/rules/0475-suricata_rules.xml`. The base alert rule (86601) is set to level 3 — below most alert thresholds. We override it with a custom rule at level 7 so it appears in the dashboard.

WARNING: Do NOT edit files under `/var/ossec/ruleset/rules/` — these are managed by Wazuh and will be overwritten on updates. Always use `/var/ossec/etc/rules/local_rules.xml` for custom rules.

Edit `local_rules.xml`:

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

Add this block inside the existing tag (or before):

```
<group name="ids,suricata,">
  <rule id="86701" level="7">
    <if_sid>86601</if_sid>
    <description>Suricata: Alert - $(alert.signature)</description>
    <group>network,ids,suricata,</group>
  </rule>
</group>
```

NOTE: Rule 86701 inherits from 86601 (the base Suricata alert rule) and raises its level to 7. The `alert.signature` field shows the exact rule name that matched, e.g. "GPL ATTACK_RESPONSE id check returned root".

Restart Wazuh Manager:

```
sudo systemctl restart wazuh-manager
```

Verify with `wazuh-logtest`:

```
sudo /var/ossec/bin/wazuh-logtest
# Paste this test JSON:
{"timestamp":"2026-06-24T12:06:14+0600","event_type":"alert",
 "src_ip":"13.35.20.90","dest_ip":"192.168.79.131",
 "alert":{"signature":"GPL ATTACK_RESPONSE id check returned root","severity":2}}
# Expected output:
# id: '86701'
# level: '7'
# Alert to be generated.
```


Step 6 — Create the Auto-Block Script on the Agent

NOTE: Run on: AGENT (rakib-virtual-machine)

This script reads the Suricata alert JSON from stdin (sent by Wazuh execd), extracts the attacker's IP, and blocks it using iptables.

WARNING: Wazuh 4.x active response scripts receive the alert JSON via stdin — NOT as command-line arguments. Scripts that read \$1/\$2/\$3/\$4 will fail with "Cannot read srcip from data". Always use read INPUT at the top of the script.

Create the script:

```
sudo nano /var/ossec/active-response/bin/suricata-drop.sh
```

Paste the complete script:

```
#!/bin/bash
read -r INPUT
SRC_IP=$(echo "$INPUT" | grep -oP '"parameters".*?"alert".*?"data".*?"src_ip": "\K[^"]+' | head -1)
LOG=/var/ossec/logs/active-responses.log
if [ -z "$SRC_IP" ]; then
    echo "$(date) - suricata-drop: Cannot find src_ip" >> $LOG
    exit 1
fi
echo "$(date) - suricata-drop: Blocking IP $SRC_IP" >> $LOG
iptables -I INPUT -s $SRC_IP -j DROP
iptables -I FORWARD -s $SRC_IP -j DROP
exit 0
```

Set correct permissions:

```
sudo chmod 750 /var/ossec/active-response/bin/suricata-drop.sh
sudo chown root:wazuh /var/ossec/active-response/bin/suricata-drop.sh
# Verify
sudo ls -la /var/ossec/active-response/bin/suricata-drop.sh
# Expected: -rwxr-x--- 1 root wazuh ... suricata-drop.sh
```

Step 7 — Configure Active Response on the Manager

NOTE: Run on: *MANAGER* (*wazuh-soc*)

Edit `ossec.conf`:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add these two blocks before :

```
<!-- Command definition – must come before active-response -->
<command>
  <name>suricata-drop</name>
  <executable>suricata-drop.sh</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>

<!-- Active response – triggers when rule 86701 fires -->
<active-response>
  <command>suricata-drop</command>
  <location>local</location>
  <rules_id>86701</rules_id>
  <timeout>600</timeout>
</active-response>
```

Settings explained:

Setting	Value	Meaning
location	local	Run on the agent where the alert originated
rules_id	86701	Only trigger for our Suricata rule
timeout	600	Auto-unblock the IP after 10 minutes

Restart Wazuh Manager:

```
sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager
```

Step 8 — Test the Full Pipeline

NOTE: Run on: AGENT (rakib-virtual-machine)

testmynids.org is a website specifically built to test Suricata/Snort detection. When you curl it, it responds with "uid=0(root) gid=0(root) groups=0(root)" — a string that matches Suricata rule 2100498 (GPL ATTACK_RESPONSE id check returned root).

Open two terminals on the agent:

Terminal 1 — watch the active response log:

```
sudo tail -f /var/ossec/logs/active-responses.log
```

Terminal 2 — send the test request:

```
curl http://testmynids.org/uid/index.html
# Expected response: uid=0(root) gid=0(root) groups=0(root)
```

Expected output in Terminal 1:

```
Wed Jun 24 12:54:56 2026 - suricata-drop: Blocking IP 13.35.20.97
```

Verify the IP is blocked in iptables:

```
sudo iptables -L INPUT | grep 13.35.20
# Expected: DROP all -- server-13-35-20-97.cloudfront.net anywhere
```

Verify alert in Wazuh Dashboard:

- Go to Security Events in the Wazuh Dashboard
- Search for: rule.groups: suricata
- You should see an alert with rule.id: 86701, level: 7
- data.alert.signature will show: GPL ATTACK_RESPONSE id check returned root
- data.src_ip will show the CloudFront IP that was blocked

Troubleshooting

suricata-update conflict during install

Cause: Suricata 8.x bundles suricata-update internally. The old standalone package conflicts during upgrade.

Fix:

```
sudo apt remove suricata-update -y
sudo apt install suricata -y
```

eve.json Permission denied for wazuh user

Cause: /var/log/suricata/ is owned by the suricata group. The wazuh user needs to be added to it.

Fix:

```
sudo usermod -aG suricata wazuh
sudo systemctl restart wazuh-agent
```

Rule 86701 shows level 3 instead of 7

Cause: The built-in rule 86601 is at level 3 and is loading before your custom rule. Make sure you are using rule ID 86701 (not 86601) in local_rules.xml.

Fix:

```
sudo /var/ossec/bin/wazuh-logtest # paste test JSON and check level
```

"Cannot read srcip from data" in active-responses.log

Cause: The old firewall-drop script is running instead of suricata-drop.sh. Check that the active-response block in ossec.conf uses suricata-drop.

Fix:

```
sudo grep -B5 "86701" /var/ossec/etc/ossec.conf
# Should show: <command>suricata-drop</command>
```

No new entries appear in active-responses.log after curl

Cause: Suricata may not have rules loaded. Check rules_loaded count.

Fix:

```
sudo suricata -T -c /etc/suricata/suricata.yaml -v 2>&1 | grep "rules successfully loaded"
# Expected: 50861 rules successfully loaded
# If 0 rules: run suricata-update first
sudo suricata-update && sudo systemctl restart suricata
```

Quick Reference

Key File Paths

File / Component	Path
Suricata main config	/etc/suricata/suricata.yaml
Suricata alert log (JSON)	/var/log/suricata/eve.json
Suricata fast alert log	/var/log/suricata/fast.log
Suricata rules directory	/var/lib/suricata/rules/
Built-in Suricata Wazuh rules	/var/ossec/ruleset/rules/0475-suricata_rules.xml
Custom Wazuh rules	/var/ossec/etc/rules/local_rules.xml
Auto-block script (on AGENT)	/var/ossec/active-response/bin/suricata-drop.sh
Active response log (on AGENT)	/var/ossec/logs/active-responses.log
Manager alert log	/var/ossec/logs/alerts/alerts.log
Agent ossec.conf	/var/ossec/etc/ossec.conf (on agent)
Manager ossec.conf	/var/ossec/etc/ossec.conf (on manager)

Useful Commands

What it does	Command
Test Suricata config	sudo suricata -T -c /etc/suricata/suricata.yaml -v
Update Suricata rules	sudo suricata-update
Watch live alerts	sudo tail -f /var/log/suricata/eve.json grep alert
Check Wazuh rule match	sudo /var/ossec/bin/wazuh-logtest
Watch auto-block log	sudo tail -f /var/ossec/logs/active-responses.log
View blocked IPs	sudo iptables -L INPUT grep DROP
Manually unblock an IP	sudo iptables -D INPUT -s <IP> -j DROP
Search Suricata alerts	sudo grep '"event_type":"alert"' /var/log/suricata/eve.json tail -5
Restart Suricata	sudo systemctl restart suricata
Restart Wazuh Manager	sudo systemctl restart wazuh-manager
Restart Wazuh Agent	sudo systemctl restart wazuh-agent

Final Setup Checklist

Go through this every time you set this up on a new server:

Task	Done?
Suricata 8.0.5 installed via OISF stable PPA	☐
suricata-update run — 50,000+ rules loaded	☐
Network interface set to correct name (ens33) in suricata.yaml	☐
eve.json output enabled with alert, http, dns, tls, flow types	☐
suricata -T config test passes with no errors	☐
Suricata service running: systemctl status suricata	☐
wazuh user added to suricata group via usermod -aG	☐
eve.json readable by wazuh user: sudo -u wazuh cat /var/log/suricata/eve.json	☐
localfile block added to agent ossec.conf for eve.json	☐
Wazuh agent log shows: Analyzing file: /var/log/suricata/eve.json	☐
Rule 86701 added to local_rules.xml on Manager at level 7	☐
wazuh-logtest confirms level 7 and "Alert to be generated"	☐
suricata-drop.sh created on AGENT with permissions 750, owner root:wazuh	☐
Manager ossec.conf has <command>suricata-drop</command> block BEFORE <active-response>	☐
curl http://testmynids.org/uid/index.html triggers alert	☐
active-responses.log shows: suricata-drop: Blocking IP ...	☐
iptables -L INPUT shows DROP rule for blocked IP	☐
Wazuh Dashboard shows alert under rule.groups: suricata	☐

-- End of Document --